

The reverse-complement quotient is a signed graph: what Zaslavsky's theory gives, and what it cannot

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Abstract

The companion note built the reverse-complement double cover $D_k(S)$ of a de Bruijn multidigraph and found that the involution $\rho(v) = \bar{v}$ carries D_k to its *opposite* digraph, so that ρ induces no endomorphism of $\mathcal{K}(D_k)$ and the cover does not descend. The obvious repair is to pass to the quotient $\Sigma_k = D_k/\rho$ and to use Zaslavsky's signed Laplacian there. Half of that works, and the half that fails is the interesting half. *Works:* ρ is free exactly when no repeated k -mer is its own reverse complement, which is automatic for odd k and genuinely fails for even k (ϕ X174 has three palindromic repeated 10-mers and one palindromic repeated 12-mer); every bundle of Σ_k is monochromatic, so Σ_k carries honest r -bundles; Σ_k is *unbalanced* precisely when S carries an inverted repeat of length $\geq k$, which is Zaslavsky balance read through the connectivity criterion of the companion note; and the Reiner–Tseng sequence $0 \rightarrow \mathcal{K}(\Sigma_k) \rightarrow \mathcal{K}(|D_k|) \rightarrow \mathcal{K}(|\Sigma_k|) \rightarrow 0$ holds on the nose, with a non-split 2-primary part. *Fails, twice.* First, if P is the permutation matrix of ρ then $P\Delta(D_k)P = \Delta(D_k)^\top$ and never $\Delta(D_k)$: the deck involution is an *anti*-automorphism of the Laplacian, the ± 1 eigenlattices of P are not Δ -stable, and the *directed* sandpile group — the one the BEST theorem counts reconstructions with — lies in no covering sequence. What it gives instead is a perfect *symmetric* linking form $\mathcal{K}(D_k) \times \mathcal{K}(D_k) \rightarrow \mathbb{Q}/\mathbb{Z}$, so the relevant notion is a metabolizer rather than a Lagrangian; and since $|\mathcal{K}(D_{11})| = 25\,318\,912$ is not a square, ϕ X174 has none. Second, a signed r -bundle chain does not contract: the presentation column at the chain's endpoint differs between Σ and Σ/C by exactly $sr d_0$, the series law, so the splitting theorem that makes the genome-scale computation feasible on one strand has no bidirected analogue. What does survive is the r -rank, and exactly so: modulo a prime p dividing r an interior vertex of the chain has degree and adjacencies all $\equiv 0$, so its row and column of the signed Laplacian vanish and Σ and Σ/C reduce to the same matrix. That one observation — a *p-divisible subgraph is invisible mod p* — proves the p -rank gain to be exactly s for every prime $p \mid r$, composite r included, and it is also what the $\mathbb{F}_7$ -nullity below measures. The seven ribosomal operons of *E. coli* K-12 merge into genuine 7-bundles of Σ_k whose longest block spans 728 bp at $k = 31, 51$ and 75 alike — the same block the companion note matched one-to-one onto the seven operons — and the only trace the quotient keeps of them is the $\mathbb{F}_7$ -nullity of the signed Laplacian.

1 What was asked, and what is true

The single-strand theory of these notes assigns to a sequence S and a word length k the de Bruijn multidigraph $G_k(S)$, whose vertices are the repeated k -mers and whose arcs are the $(k+1)$ -mer

occurrences, and reads the reconstruction count off the critical group $\mathcal{K}(G_k)$ through the BEST theorem. Double-strandedness was handled in the companion note by passing to the *double cover* $D_k(S)$: the multidigraph whose arcs are the $(k+1)$ -mer occurrences in S and in $\bar{S}$. That graph is balanced, the unitig lemma and the splitting theorem apply to it verbatim, and the seven ribosomal operons of *E. coli* — five on one strand, two on the other — finally appear as one seven-copy family instead of a five-copy and a two-copy family.

What the companion note could not do was quotient. The map $\rho(v) = \bar{v}$ satisfies

$$\rho: D_k \longrightarrow D_k^{\text{op}}, \quad \rho(u \rightarrow v) = \bar{v} \rightarrow \bar{u},$$

so it reverses arcs; the quotient of a digraph by a direction-reversing involution is not a digraph, and ρ induces a map $\mathcal{K}(D_k) \rightarrow \mathcal{K}(D_k^{\text{op}}) \cong \mathcal{K}(D_k)^\vee$ rather than an endomorphism. The standard repair in combinatorics is to forget the directions, to record instead a *sign* on each edge of the quotient, and to work with Zaslavsky’s signed Laplacian $\Delta_\Sigma = D - A_\sigma$ [7]. This note carries that out, and reports both what it yields and the two places where it provably cannot yield what was wanted.

The one-paragraph summary is this. The quotient is a bona fide signed graph, its critical group is a switching invariant of the genome, its balance has an exact biological meaning, and the integral covering theory of Reiner and Tseng [6] applies to it word for word. But that theory computes an invariant of the *undirected shadow* $|D_k|$, which by the matrix-tree theorem counts spanning trees; the quantity of interest counts Eulerian circuits and lives on the digraph, and Theorem 9 says the digraph admits no such theory. Separately, Theorem 11 says the chain contraction that reduces a bacterial genome from tens of thousands of vertices to a few hundred has no signed analogue. Together these explain why the “unified bidirected sandpile group” that one would like to write down does not exist, and what the correct residue of the seven operons is instead.

Throughout, S is a circular sequence of length N over $\{A, C, G, T\}$, $\bar{S}$ its reverse complement, and $\bar{w}$ the reverse complement of a word w . All computations are exact integer arithmetic; the script `bidirected_sandpile.py` reproduces every number below and prints a seven-line pass/fail battery.

2 The quotient, and when it exists

Definition 1. $D_k = D_k(S)$ is the multidigraph of the companion note: one vertex per k -mer occurring at least twice in the concatenation $S\bar{S}$, one arc $u \rightarrow v$ per occurrence of a $(k+1)$ -mer w with prefix u and suffix v in S or in $\bar{S}$. The involution ρ sends the vertex w to $\bar{w}$ and the arc $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$.

The first question is whether ρ acts freely on vertices, because the whole signed-graph dictionary presupposes a free $\mathbb{Z}/2$ action.

Proposition 2. ρ fixes a vertex of D_k if and only if some repeated k -mer w satisfies $\bar{w} = w$. For odd k no such word exists over any alphabet with a fixed-point-free complement, so ρ is free. For even k such words exist and do occur.

Proof. A word w of odd length k with $\bar{w} = w$ would have its middle letter equal to its own complement, and complementation on $\{A, C, G, T\}$ has no fixed point. For even k the condition $\bar{w} = w$ is the condition that w be a palindromic restriction-site-like word, and these are common. $\square$

k	parity	$ V(D_k) $	ρ -fixed	A_σ symmetric
9	odd	396	0	yes
10	even	125	3	no
11	odd	30	0	yes
12	even	7	1	no

Table 1: $\phi X174$ (NC_001422.1). Palindromic repeated k -mers exist at $k = 10$ and $k = 12$, and at exactly those k the quotient adjacency matrix fails to be symmetric, so no signed graph is defined. Check (P).

Table 1 makes the point concretely and shows that the obstruction is not hypothetical: it is precisely at the even k with palindromic repeats that the construction below breaks, and it breaks visibly, in the symmetry of the adjacency matrix. From here on k is odd, or even with no palindromic repeated k -mer.

Remark 3 (What the fixed points are, topologically). The failure at even k is not a technicality but a change of category. A vertex $w = \bar{w}$ is a branch point of the projection $D_k \rightarrow D_k/\rho$, so the quotient is not a covering space and the fibre over w is a single point rather than two. In Zaslavsky’s language the image acquires a *half-edge* — an edge with one endpoint — at w : an arc $u \rightarrow w$ and its partner $w \rightarrow \bar{u}$ descend to the same edge of the quotient but attach to $[w]$ with opposite end-signs, since $\varepsilon(w) = +1$ forces the two signs $\varepsilon(u)\varepsilon(w)$ and $\varepsilon(w)\varepsilon(\bar{u})$ to differ. That is exactly why the quotient adjacency matrix in Table 1 fails to be symmetric: A_σ records one sign per edge, and a half-edge has two. Reiner and Tseng do treat signed graphs with half-edges [6, §9], so the object is not outside the theory, but its critical group is no longer the one computed below, and the double cover it belongs to is branched. Biologically the fixed points are the palindromic restriction-site-like words, so the branch locus of the quotient is precisely the repeated restriction sites of the genome.

Definition 4. Choose a representative of each ρ -orbit $\{v, \bar{v}\}$ and set $\varepsilon(v) = +1$ if v is the chosen one and -1 otherwise. The signed graph $\Sigma_k = D_k/\rho$ has one vertex per ρ -orbit of vertices, one edge per ρ -orbit of arcs, and the edge coming from $u \rightarrow v$ carries the sign $\varepsilon(u)\varepsilon(v)$.

The arcs do fall into orbits of size two, and here no parity hypothesis is needed. An arc of D_k is an *occurrence*: a position in S or a position in $\bar{S}$. The involution ρ carries occurrences in S to occurrences in $\bar{S}$ and back, and no occurrence lies in both, so ρ is free on arcs outright. The script nevertheless constructs the pairing explicitly and fails loudly if any arc lacks a partner.

Two facts make Definition 4 usable.

Lemma 5. *Every bundle of parallel edges of Σ_k is monochromatic.*

Proof. The sign $\varepsilon(u)\varepsilon(v)$ depends only on the endpoints of the arc, not on the arc. All r arcs $u \rightarrow v$ therefore descend to r edges of the same sign. $\square$

This is what licenses the phrase “ r -bundle” below, and it is not automatic for a general signed multigraph.

Lemma 6. *Changing the choice of representatives is a Zaslavsky switching, and Δ_{Σ_k} changes by conjugation by a diagonal ± 1 matrix. Hence $\text{coker } \Delta_{\Sigma_k}$, and with it the critical group of Section 4, depends only on (S, k) .*

Replacing the representative at one orbit multiplies ε by -1 there, which multiplies the corresponding row and column of A_σ by -1 and leaves D alone. The script verifies invariance directly on eight random representative sets at $k = 9$ and $k = 11$, obtaining for instance

$$\mathcal{K}(\Sigma_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481$$

every time. Check (W).

3 Balance is an inverted repeat

Theorem 7. Σ_k is unbalanced if and only if S carries an inverted repeat of length at least k .

Proof. A signed graph is balanced if and only if its signed double cover is disconnected; for the $\mathbb{Z}/2$ -voltage double cover this is Reiner and Tseng, Propositions 11.4 and 11.5 [6]. The underlying multigraph of D_k is that double cover, by Definition 4. The companion note proves that D_k is connected exactly when S carries an inverted repeat of length at least k , the two components otherwise being the plus-strand and minus-strand halves. $\square$

For $\phi X174$ the longest inverted repeat has length 12, realised by ATTAAGCTCATT at position 441 against its reverse complement at 1448. The script confirms the trichotomy at $k = 9, 11, 13, 15$: unbalanced and connected at 9 and 11, balanced and disconnected at 13 and 15. Check (B).

Theorem 7 is worth stating because it converts a purely combinatorial dichotomy into a statement about the sequence. A genome with no inverted repeat as long as k gives a balanced Σ_k ; balanced means switching-equivalent to all-positive, which means the signs carry no information, which means the whole construction has collapsed back to the single-strand one. The sign structure is *exactly* the record of inverted repeats, nothing more and nothing less.

4 The covering theory that does exist

Reiner and Tseng define the critical group of a signed graph not as $\text{coker } \Delta_\Sigma$ but as $\mathcal{K}(\Sigma) = \text{im } \partial / \text{im } \Delta_\Sigma$, where ∂ is Zaslavsky's signed incidence matrix [6, eq. (1.3), §9]. For a connected unbalanced Σ one has $\text{im } \partial = \{x \in \mathbb{Z}^V : \sum_v x_v \text{ even}\}$, an index-two sublattice of $\mathbb{Z}^V$, so $\mathcal{K}(\Sigma)$ is an index-two subgroup of $\text{coker } \Delta_\Sigma$; ignoring the distinction doubles every order and corrupts the 2-primary part. (The competing convention of Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [3] deletes a sink row and column instead, which is the chip-firing-natural choice but a different group.) We use the Reiner–Tseng convention, computed from the explicit basis $2e_0, e_0 + e_1, \dots, e_0 + e_{n-1}$ of $\text{im } \partial$.

Theorem 8 (Reiner–Tseng [6], Thm. 1.2, verified here). *For every odd k with Σ_k unbalanced there is a short exact sequence*

$$0 \longrightarrow \mathcal{K}(\Sigma_k) \longrightarrow \mathcal{K}(|D_k|) \longrightarrow \mathcal{K}(|\Sigma_k|) \longrightarrow 0,$$

split on p -primary components for odd p and in general not split at $p = 2$.

The $\phi X174$ instance at $k = 11$ is a textbook illustration:

$$\begin{aligned} \mathcal{K}(\Sigma_{11}) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481, & \text{order } 16\,817\,924, \\ \mathcal{K}(|D_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757 \oplus \mathbb{Z}/4204481, & \text{order } 19\,738\,255\,595\,056, \\ \mathcal{K}(|\Sigma_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757, & \text{order } 1\,173\,644. \end{aligned}$$

The orders multiply exactly. The odd parts split visibly: 4204481 comes from the signed graph, $23 \cdot 12757$ from the base. The 2-primary parts do *not*: an extension of $(\mathbb{Z}/2)^2$ by $(\mathbb{Z}/2)^2$ has produced $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4$, of the right order 16 and the wrong isomorphism type. This is precisely the failure Reiner and Tseng exhibit with $0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 0$, arising here from a real bacteriophage. Check (R).

5 The deck involution is an anti-automorphism

Theorem 8 is a statement about $|D_k|$, the undirected shadow. The quantity the single-strand theory actually computes is $\mathcal{K}(D_k)$, the sandpile group of the *digraph*, because that is what the BEST theorem multiplies by $\prod_v (d^+(v) - 1)!$ to count reconstructions. The following says the two are unrelated, and says why.

Theorem 9. *Let $\Delta = \Delta(D_k)$ be the out-degree Laplacian of D_k and P the permutation matrix of ρ . Then*

$$P \Delta P = \Delta^\top,$$

and $P \Delta P \neq \Delta$ whenever D_k has an arc that is not part of a ρ -symmetric digon. Consequently Δ does not commute with P , the ± 1 eigenlattices of P are not Δ -stable, and there is no decomposition of Δ over $\mathbb{Z}$, or over $\mathbb{Z}[1/2]$, or over any ring, into a part on the invariants and a part on the anti-invariants.

Proof. ρ is a bijection from the arc set of D_k to itself carrying $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$. Hence the adjacency matrix satisfies $A_{\bar{v}\bar{u}} = A_{uv}$, that is $PAP = A^\top$, and $d^+(\bar{v}) = d^-(v) = d^+(v)$ since D_k is balanced, so $PDP = D$. Subtracting gives $P \Delta P = \Delta^\top$. If $P \Delta P$ were also Δ then $\Delta = \Delta^\top$, forcing $A = A^\top$, i.e. every arc paired with a reverse arc of the same multiplicity.

For the consequence: a decomposition along the eigenlattices of P requires $\Delta P = P \Delta$. Here $\Delta P = P \Delta^\top$, and $\Delta^\top \neq \Delta$. $\square$

This is the structural reason the companion note's obstruction was not an accident of its proof. The classical covering-graph factorisation — of Bilu and Linial [1], in the form used for signed graphs — decomposes the Laplacian of a double cover into an unsigned and a signed part on the ± 1 eigenspaces of the deck transformation. That decomposition needs the deck transformation to *commute* with the Laplacian. Here it conjugates the Laplacian to its transpose instead, and the two eigenlattices are exchanged rather than preserved.

The numbers are unambiguous. For $\phi X174$ at $k = 11$,

$$|\mathcal{K}(D_{11})| = 25\,318\,912, \quad |\mathcal{K}(|D_{11}|)| = 19\,738\,255\,595\,056,$$

and the second is not a multiple of the first; at $k = 9$ the two orders are 124- and 205-digit integers, again with no divisibility. The directed group is not an extension of anything by $\mathcal{K}(\Sigma_k)$ and is not a quotient of anything by it. Check (N).

Remark 10 (What ρ gives instead: a symmetric linking form). The honest formulation is that ρ makes D_k into a digraph with a *duality*, not a digraph with a symmetry. Bowen–Franks duality (Paper VIII, Lem. 1.5 of the main series) says coker M and coker $M^\top$ are canonically dual for a square integer matrix M with finite cokernel, and Theorem 9 identifies $\Delta^\top$ with $P \Delta P$. Composing gives a perfect pairing on the sandpile group itself:

$$\langle u, v \rangle = (Pu)^\top \Delta^{-\top} v \bmod \mathbb{Z}, \quad \mathcal{K}(D_k) \times \mathcal{K}(D_k) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

It is well defined — if $u = \Delta^\top z$ then $Pu = P\Delta^\top P(Pz) = \Delta(Pz)$, so $\langle u, v \rangle = (Pz)^\top v \in \mathbb{Z}$ — and it is *symmetric*, not alternating: inverting $P\Delta P = \Delta^\top$ gives $P\Delta^{-\top} = \Delta^{-1}P$, whence $\langle u, v \rangle = u^\top P\Delta^{-\top}v = u^\top \Delta^{-1}Pv = \langle v, u \rangle$. Verified exactly for $\phi X174$ at $k = 11$ and $k = 12$; check (D).

This matters for how one should look for a double-stranded count. A symmetric linking form has no Lagrangians in the symplectic sense; the relevant notion is a *metabolizer*, a subgroup L with $L = L^\perp$, and a metabolizer forces $|\mathcal{K}(D_k)| = |L|^2$. For $\phi X174$ that already fails: $|\mathcal{K}(D_{12})| = 33$ and $|\mathcal{K}(D_{11})| = 25\,318\,912$, neither a square. So whatever counts double-stranded reconstructions, it is not the order of a self-orthogonal subgroup of $\mathcal{K}(D_k)$, and the symmetry of the form — not merely its existence — is the first thing a candidate theory has to respect.

One further negative is worth recording, because it removes the most natural-looking formulation. The sandpile group is a torsor on the arborescences *converging* on a fixed root, and those are what the BEST theorem counts with. One would like to say that a double-stranded count is a sum over the ρ -orbits of arborescences. There are none: ρ reverses arcs, so it carries an arborescence converging on r to one *diverging* from $\rho(r)$, and the two families are disjoint. What ρ gives at tree level is therefore a *bijection*

$$\rho: \{\text{arborescences converging on } r\} \xrightarrow{\sim} \{\text{arborescences diverging from } \rho(r)\},$$

both families having $|\mathcal{K}(D_k)|$ elements — the tree-level shadow of the duality, not an action. Verified for $\phi X174$ at $k = 12$ at three roots, where each family has 33 members and the image lands in the diverging family every time and in the converging one never; check (D). A double-stranded theory must be built from that bijection.

6 Signed bundle chains do not contract

The single-strand computation for a bacterial genome is only possible because of the splitting theorem: a *chain* $C = (c_1, \dots, c_s)$ of vertices each of which sends all r of its out-arcs to the next splits off a factor $(\mathbb{Z}/r)^{s-1}$ and can be contracted, reducing *E. coli* at $k = 75$ from 28 579 vertices to a skeleton of 261. The question is whether this survives in Σ_k . It does not, and the reason is elementary enough to state exactly.

Theorem 11. *Let Σ be a signed graph containing a chain $C = (c_1, \dots, c_s)$: vertices of degree exactly $2r$, joined by r -bundles to c_{t-1} and c_{t+1} , where $c_0 = a$ and $c_{s+1} = b$ lie outside C . Switch so that every chain bundle is positive, and write $d_t = x_{c_t} - x_{c_{t+1}}$ for $t = 0, \dots, s$ in coker Δ_Σ , where x_v is the class of the basis vector at v . Then*

- (a) *the presentation columns at $c_1, \dots, c_s$ say exactly $r(d_{t-1} - d_t) = 0$ for $t = 1, \dots, s$; in particular the s second differences $\delta_t = x_{c_{t-1}} - 2x_{c_t} + x_{c_{t+1}}$ are killed by r , and $rd_0 = rd_1 = \dots = rd_s$;*
- (b) *the column at a in Σ and the column at a in the contraction Σ/C — which replaces C by a single r -bundle $a-b$ of the product sign — differ by exactly $sr d_0$.*

Consequently $\mathcal{K}(\Sigma) \not\cong (\mathbb{Z}/r)^s \oplus \mathcal{K}(\Sigma/C)$ in general.

Proof. (a) Δ_Σ is symmetric and its column at c_t has entry $2r$ at c_t and $-r$ at each of c_{t-1}, c_{t+1} , all other entries zero. The relation it imposes is $2r x_{c_t} - r x_{c_{t-1}} - r x_{c_{t+1}} = 0$, which is $-r(d_{t-1} - d_t) = 0$. Chaining the s relations gives $rd_0 = \dots = rd_s$.

r	s	$\mathcal{K}(\Sigma)$	$\mathcal{K}(\Sigma/C)$	ratio	r^s
3	1	$(\mathbb{Z}/3)^2 \oplus \mathbb{Z}/8$	$\mathbb{Z}/2 \oplus \mathbb{Z}/9$	4.00	3
3	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^3 \oplus \mathbb{Z}/8$	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/9$	12.00	9
3	3	$(\mathbb{Z}/3)^3 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/9$	$\mathbb{Z}/2 \oplus \mathbb{Z}/9$	54.00	27
5	2	$(\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/5)^2 \oplus \mathbb{Z}/17$	$\mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/7$	30.36	25
5	3	$\mathbb{Z}/4 \oplus (\mathbb{Z}/5)^3 \oplus \mathbb{Z}/11$	$\mathbb{Z}/2 \oplus \mathbb{Z}/13$	211.54	125
7	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/7)^2 \oplus \mathbb{Z}/23$	$\mathbb{Z}/2 \oplus \mathbb{Z}/17$	66.29	49
7	3	$\mathbb{Z}/4 \oplus (\mathbb{Z}/7)^3 \oplus \mathbb{Z}/13$	$\mathbb{Z}/2 \oplus \mathbb{Z}/17$	524.59	343
11	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/11)^2 \oplus \mathbb{Z}/64$	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/29$	133.52	121

Table 2: An r -bundle chain of s internal vertices grafted into a fixed unbalanced base. The r -rank rises by exactly s in all eight rows — the $(\mathbb{Z}/r)^s$ of Theorem 11(a) is visible, and Theorem 13 says it must be exactly s — but the order ratio never equals r^s , and is not even an integer in five rows. Check (C); check (Q) extends the p -rank column to composite $r = 4, 6, 9, 10, 12, 15, 21, 25$ and to $p = 2$, and to p -divisible trees, cycles and multi-attachment subgraphs.

(b) The column at a in Σ contains $-r x_{c_1}$ where the column at a in Σ/C contains $-r x_b$; all other entries agree, since $\deg a$ is r in both. The difference is $r(x_b - x_{c_1}) = -r \sum_{t=1}^s d_t$, and by (a) each rd_t equals rd_0 , so this is $-sr d_0$. $\square$

Contrast the directed case. There the column at c_t is $r(x_{c_t} - x_{c_{t+1}}) = rd_t$, which kills each *first* difference outright; the chain contributes $(\mathbb{Z}/r)^{s-1}$ as a direct summand and contracts cleanly. In the signed case only second differences die. The residual first difference d_0 is not torsion; it is coupled to the rest of the graph, and the coupling is the series law of electrical networks — $s+1$ bundles of conductance r in series behave like one bundle of conductance $r/(s+1)$, which is not an integer multiplicity, so there is no contracted signed graph to compare to.

Table 2 separates the two halves of the theorem. What does not survive contraction is the order and the isomorphism type. What survives is the r -rank, and the reason is a one-line observation that has nothing to do with chains.

Lemma 12 (p -divisible subgraphs are invisible mod p). *Let Σ be a signed multigraph, p a prime, and H a set of bundles of Σ every one of which has multiplicity divisible by p . Then*

$$\text{rank}_{\mathbb{F}_p} \Delta_\Sigma = \text{rank}_{\mathbb{F}_p} \Delta_{\Sigma \setminus H},$$

where $\Sigma \setminus H$ is Σ with the bundles of H deleted and the vertex set kept.

Proof. Call a vertex *interior* to H if every bundle at it lies in H . Its degree is a sum of multiplicities divisible by p , hence $\equiv 0$, and each of its adjacency entries is $\pm$ a multiplicity in H , hence $\equiv 0$: its whole row and column of Δ_Σ vanish mod p , and they vanish in $\Delta_{\Sigma \setminus H}$ too, that vertex now being isolated. A vertex not interior to H loses, on passing to $\Sigma \setminus H$, exactly the H -part of its degree and its H -adjacencies; both are $\equiv 0$. So the two matrices are congruent mod p entry by entry. $\square$

Theorem 13 (the r -rank gain, exactly). *Let Σ be connected and unbalanced with an r -bundle chain C of s internal vertices, and let Σ/C be connected and unbalanced as well. Then for every prime p dividing r ,*

$$p\text{-rank}(\text{coker } \Delta_\Sigma) - p\text{-rank}(\text{coker } \Delta_{\Sigma/C}) = s,$$

k	$ V(D_k) $	$ V(\Sigma_k) $	ρ -fixed	7-bundles	longest 7-chain	span (bp)	$\dim_{\mathbb{F}_7} \ker \Delta_{\Sigma_k}$
31	60 548	30 274	0	4919	698	728	7441
51	53 380	26 690	0	4352	678	728	6581
75	48 498	24 249	0	3814	654	728	5864

Table 3: *E. coli* K-12 MG1655, NC_000913.3, 4 641 652 bp, at the three odd word lengths of the single-strand study. The longest 7-bundle chain spans the same 728 base pairs at all three k — $698 + 31 - 1 = 678 + 51 - 1 = 654 + 75 - 1$ — which is the seven-copy block the companion note matched one-to-one onto the seven ribosomal operons. Check (E).

and for every odd prime $p \mid r$ the same holds with $\text{coker } \Delta$ replaced by the Reiner–Tseng group $\mathcal{K}$. More generally, if H is any p -divisible subgraph with w interior vertices then the p -rank of $\text{coker } \Delta$ exceeds that of $\text{coker } \Delta_{\Sigma \setminus H}$ by exactly w .

Proof. Both the chain’s bundles in Σ and the single a – b bundle of Σ/C have multiplicity r , so Lemma 12 applied to each deletes them and leaves the same matrix $\Delta_{\Sigma'}$, Σ' being Σ with the chain and both end bundles removed. Hence $\text{rank}_{\mathbb{F}_p} \Delta_{\Sigma} = \text{rank}_{\mathbb{F}_p} \Delta_{\Sigma/C}$, while Δ_{Σ} is larger by the s interior vertices, so its nullity mod p exceeds that of $\Delta_{\Sigma/C}$ by s . For a connected unbalanced signed graph Δ is positive definite, hence nonsingular, so $\text{coker } \Delta$ is finite and its p -rank — the number of invariant factors divisible by p — equals $\dim_{\mathbb{F}_p} \text{coker } \Delta \otimes \mathbb{F}_p = \text{nullity}_{\mathbb{F}_p} \Delta$. For odd p , $\mathcal{K}(\Sigma)$ has index two in $\text{coker } \Delta_{\Sigma}$ and therefore the same p -primary part. The general statement is the same argument with s replaced by the number of interior vertices of H . $\square$

Remark 14 (what happens at $p = 2$). For $p = 2$ the index-two subgroup does matter. Tensoring $0 \rightarrow \mathcal{K}(\Sigma) \rightarrow \text{coker } \Delta_{\Sigma} \rightarrow \mathbb{Z}/2 \rightarrow 0$ with $\mathbb{F}_2$ gives 2-rank $\mathcal{K} = 2\text{-rank } \text{coker } \Delta - 1 + \varepsilon$ with $\varepsilon \in \{0, 1\}$ recording whether the extension splits at 2, so the gain for $\mathcal{K}$ is $s + \varepsilon_{\Sigma} - \varepsilon_{\Sigma/C}$. In the basis $2e_0, e_0 + e_1, \dots, e_0 + e_{n-1}$ of $\text{im } \partial$ the chain’s column reduces mod 2 to a single possibly nonzero entry in the row of $2e_0$, which bounds the deviation by one. In 5828 connected unbalanced instances with even r the two ε agree and the gain is s ; that they always agree is *not* proved.

Theorem 13 is a partial and negative data point for Question 5.5 of [3], which asks how $\mathcal{K}(\Sigma_{\varphi})$ depends on $\mathcal{K}(|\Sigma|)$ as the signature varies and observes only that the number of invariant factors seems to be preserved: here the count of factors divisible by p is pinned exactly, while the order and the isomorphism type are not.

Corollary 15. *The reduction that makes $\mathcal{K}(G_k)$ computable for a bacterial genome — contract every maximal chain, compute on the skeleton, multiply back — has no analogue for $\mathcal{K}(\Sigma_k)$.*

7 *E. coli* K-12: what is left of the seven operons

The construction does what was asked of it structurally. ρ is free at all three odd word lengths, Σ_k is a signed graph, and the seven operons appear as genuine 7-bundles: 4919 of them at $k = 31$, organised into chains of which the longest has 698 vertices. That chain spans $698 + 31 - 1 = 728$ base pairs, and the corresponding numbers at $k = 51$ and $k = 75$ give 728 as well. The agreement with the companion note is literal and not merely numerical: working on D_{51} rather than on Σ_{51} , and identifying the chain’s seven occurrences against the RefSeq annotation, it reports a longest multiplicity-seven chain of $s = 678$ spanning 728 bp whose occurrences hit the seven `rrn`

operons one each. The quotient neither shortens that chain nor merges it with its own reverse complement, so 678 appears again here. On the structural question — do the five plus-strand and two minus-strand copies become one seven-copy object in the quotient — the answer is yes.

On the quantitative question the answer is no, for the two reasons already proved. Corollary 15 says $\mathcal{K}(\Sigma_k)$ cannot be computed by contraction at this size, and Theorem 9 says that even if it could, it would not count reconstructions. What is computable, and what is the right residue of the operons, is the $\mathbb{F}_7$ -nullity of the signed Laplacian. The reason it is the right residue is Lemma 12 at $p = 7$: the union of all 7-bundles is a 7-divisible subgraph, hence invisible over $\mathbb{F}_7$, and every vertex interior to it — every vertex whose bundles are all 7-bundles — contributes a zero row and a unit of nullity. So the $\mathbb{F}_7$ -nullity and the exactness of the chain’s 7-rank gain are two readings of one lemma, and the numbers below are an instance of Theorem 13 at genome scale. The $\mathbb{F}_7$ -nullity therefore measures the total length of the seven-copy repeat structure, and decays from 7441 to 5864 as k rises from 31 to 75 and short seven-copy blocks fall below the word length. The corresponding single-strand figures are 3496, 3201 and 2996: roughly half, as one expects from a construction that cannot see five copies and two copies as seven.

8 Scope, provenance, and what is open

Scope. Everything proved here concerns the *quotient* Σ_k and its undirected shadow. Proposition 2, Lemma 5, Lemma 6, Theorem 9 and Theorem 11 are proved in full generality. Theorem 7 is a combination of Reiner–Tseng Prop. 11.4–11.5 with the connectivity criterion of the companion note. Theorem 8 is *not* ours: it is Reiner and Tseng’s Theorem 1.2, and what is new here is only that it is exhibited on a reverse-complement de Bruijn graph, including a non-split 2-primary extension arising from a real genome. Lemma 12 and Theorem 13 are proved in full; an earlier version of this note had the exactness of the r -rank gain as a computational observation only. Lemma 12 is elementary enough that we would not be surprised to find it in the literature on p -ranks of critical groups, and we claim novelty only for its two consequences here. What remains unproved is the case $p = 2$ for the Reiner–Tseng group: by Remark 14 the gain is $s + \varepsilon_\Sigma - \varepsilon_{\Sigma/C}$, and that the two ε always agree is verified on 5828 instances and not proved.

Provenance. The signed-graph formalism, the signed incidence and Laplacian matrices, balance and switching are Zaslavsky’s [7]; the dictionary between bidirected graphs and oriented signed graphs is Edmonds’, as recorded in Zaslavsky’s glossary [8]. The critical group of a signed graph and the covering sequence used in Section 4 are Reiner and Tseng’s [6]; the competing rooted convention and the chip-firing model for signed graphs are due to Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [3], building on the invertible-matrix chip-firing of Guzmán and Klivans [4]. The spectral form of the double-cover decomposition, which Theorem 9 shows is unavailable here, is Bilu and Linial’s [1]. The bidirected-graph model of double-stranded assembly is Medvedev, Georgiou, Myers and Brudno’s [5]. Sandpile groups of ordinary de Bruijn digraphs are treated by Chan, Hollmann and Pasechnik [2]. We are not aware of prior work computing critical groups of reverse-complement de Bruijn graphs, signed or otherwise; a targeted search found none.

Open. Three things. *First*, the residue of Theorem 13 at the prime two: show that the index-two extension of Remark 14 splits, or fails to split, for Σ and Σ/C together, so that the gain for the Reiner–Tseng group is s there as well — or exhibit the pair where it is $s \pm 1$. *Second*, the series law suggests working with weighted signed graphs over $\mathbb{Z}[1/(s+1)]$, where the contraction

does exist; whether the resulting group has any combinatorial meaning — whether it counts anything — is unknown to us. *Third*, and most important, Theorem 9 closes the quotient route to a double-stranded reconstruction count but does not close the problem. What survives is the symmetric linking form of Remark 10, and that is the structure a correct double-stranded theory should use. Two things are already known about it and both are constraints rather than encouragements: the form is symmetric, so there are no Lagrangians in the symplectic sense; and for $\phi X174$ the group order is not a square, so there is no metabolizer either. A count modulo strand exchange must therefore be a weighted sum over isotropic subgroups, or over the orbits of the form’s isometry group, rather than the order of a single distinguished subgroup. We have not attempted it, and we note that the naive guess — that the answer is $\sqrt{|\mathcal{K}(D_k)|}$ — is already excluded.

References

- [1] Y. Bilu and N. Linial, *Lifts, discrepancy and nearly optimal spectral gap*, *Combinatorica* **26** (2006), no. 5, 495–519.
- [2] S. H. Chan, H. D. L. Hollmann and D. V. Pasechnik, *Sandpile groups of generalized de Bruijn and Kautz graphs and circulant matrices over finite fields*, *J. Algebra* (2015), 268–295; arXiv:1405.0113.
- [3] M. Cho, A. Dochtermann, R. Inagaki, S. Oh, D. Snustad and B. Zacovic, *Chip-firing and critical groups of signed graphs*, preprint, arXiv:2306.09315v2 (2024).
- [4] J. Guzmán and C. Klivans, *Chip firing on general invertible matrices*, *SIAM J. Discrete Math.* **30** (2016), no. 2, 1115–1127.
- [5] P. Medvedev, K. Georgiou, G. Myers and M. Brudno, *Computability of models for sequence assembly*, in: *Algorithms in Bioinformatics (WABI 2007)*, *Lecture Notes in Computer Science*, Springer, 2007, pp. 289–301.
- [6] V. Reiner and D. Tseng, *Critical groups of covering, voltage and signed graphs*, *Discrete Math.* **318** (2014), 10–40; arXiv:1301.2977.
- [7] T. Zaslavsky, *Signed graphs*, *Discrete Appl. Math.* **4** (1982), no. 1, 47–74; erratum **5** (1983), 248.
- [8] T. Zaslavsky, *Glossary of signed and gain graphs and allied areas*, *Electron. J. Combin., Dynamic Survey #DS9* (1998).